

rmarkdown :: CHEATSHEET

What is rmarkdown?

- .Rmd files** • Develop your code and ideas side-by-side in a single document. Run code as individual chunks or as an entire document.
- Dynamic Documents** • Knit together plots, tables, and results with narrative text. Render to a variety of formats like HTML, PDF, MS Word, or MS Powerpoint.
- Reproducible Research** • Upload, link to, or attach your report to share. Anyone can read or run your code to reproduce your work.

Workflow

- 1 Open a **new .Rmd file** in the RStudio IDE by going to **File > New File > R Markdown**.
- 2 **Embed code** in chunks. Run code by line, by chunk, or all at once.
- 3 **Write text** and add tables, figures, images, and citations. Format with Markdown syntax or the RStudio Visual Markdown Editor.
- 4 **Set output format(s) and options** in the YAML header. Customize themes or add parameters to execute or add interactivity with Shiny.
- 5 **Save and render** the whole document. Knit periodically to preview your work as you write.
- 6 **Share your work!**

Embed Code with knitr

CODE CHUNKS

Surround code chunks with ``r`` and ``` or use the Insert Code Chunk button. Add a chunk label and/or chunk options inside the curly braces after `r`.

```
`r`{r chunk-label, include=FALSE}
summary(mtcars)
`r`
```

SET GLOBAL OPTIONS

Set options for the entire document in the first chunk.

```
`r`{r include=FALSE}
knitr::opts_chunk$set(message = FALSE)
`r`
```

INLINE CODE

Insert ``r` <code>` into text sections. Code is evaluated at render and results appear as text.

"Built with ``r` getRversion()`" --> "Built with 4.1.0"



SOURCE EDITOR

1. New File

2. Embed Code

3. Write Text

4. Set Output Format(s) and Options

5. Save and Render

6. Share

set preview location

insert code chunk

go to code chunk

run code chunk(s)

show outline

modify chunk options

run all previous chunks

run current chunk

VISUAL EDITOR

add/edit attributes

style options

insert citations

OPTION	DEFAULT	EFFECTS
echo	TRUE	display code in output document
error	FALSE	TRUE (display error messages in doc) FALSE (stop render when error occurs)
eval	TRUE	run code in chunk
include	TRUE	include chunk in doc after running
message	TRUE	display code messages in document
warning	TRUE	display code warnings in document
results	"markup"	"asis" (passthrough results) "hide" (don't display results) "hold" (put all results below all code)
fig.align	"default"	"left", "right", or "center"
fig.alt	NULL	alt text for a figure
fig.cap	NULL	figure caption as a character string
fig.path	"figure/"	prefix for generating figure file paths
fig.width & fig.height	7	plot dimensions in inches
collapse	FALSE	rescales output width, e.g. "75%", "300px"
comment	FALSE	collapse all sources & output into a single block
child	TRUE	prefix for each line of results
purl	TRUE	include or exclude a code chunk when extracting source code with knitr::purl()

See more options and defaults by running `str(knitr::opts_chunk$get())`

RENDERED OUTPUT

file path to output document

find in document

publish to [rpubs.com](#), [shinyapps.io](#), Posit Connect

reload document

Document Title

Author Name

- R Markdown
- Including Plots

R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

summary(mtcars)

	speed	dist
## Min.:	4.0	2.00
## 1st Qu.:	12.0	26.00
## Median:	15.0	36.00
## Mean:	15.4	42.98
## 2nd Qu.:	19.0	56.00
## Max.:	25.0	120.00

Insert Citations

Create citations from a bibliography file, a Zotero library, or from DOI references.

BUILD YOUR BIBLIOGRAPHY

- Add BibTeX or CSL bibliographies to the YAML header.

```
title: "My Document"
bibliography: references.bib
link-citations: TRUE
```

- If Zotero is installed locally, your main library will automatically be available.
- Add citations by DOI by searching "from DOI" in the **Insert Citation** dialog.

INSERT CITATIONS

- Access the **Insert Citations** dialog in the Visual Editor by clicking the **@** symbol in the toolbar or by clicking **Insert > Citation**.
- Add citations with markdown syntax by typing `[@cite]` or `@cite`.

Insert Tables

Output data frames as tables using `kable(data, caption)`.

```
`r`{r}
data <- faithful[1:4, ]
knitr::kable(data,
  caption = "Table with kable")
`r`
```

3.660	76
1.880	64
0.333	34
0.280	62

Other table packages include **flextable**, **gt**, and **kableExtra**.

Write with Markdown

The syntax on the left renders as the output on the right.

- Plain text.
- End a line with two spaces to start a new paragraph.
- Also end with a backslash \ to make a new line.
- italics* and **bold**
- superscript²/subscript₂
- ~~strikethrough~~
- escaped: * _ \\\
- endash: --, emdash: ---

- # Header 1
- ## Header 2
- ...
- ##### Header 6
- unordered list
- item 2
- item 2a (indent 1 tab)
- item 2b
1. ordered list
2. item 2
- item 2a (indent 1 tab)
- item 2b

- <link url>
- [This is a link.](link url)
- [This is another link.][id].
- At the end of the document:
- [id]: link url
- ![Caption](image.png)
- or ![Caption][id2]
- At the end of the document:
- [id2]: image.png
- http://www.posit.co/
- This is a link.
- This is another link.
- 
- Caption.

- `verbatim code`
- multiple lines of verbatim code
- > block quotes
- equation: $e^{i\pi} + 1 = 0$
- equation block:
- $$E = mc^2$$
- horizontal rule:
-

Right	Left	Default	Center
12	12	12	12
123	123	123	123
1	1	1	1

- HTML Tabsets**
- ```
Results {tabset}
Plots
text

Tables
more text
```
- Results
- Plots
- Tables
- text



# Set Output Formats and their Options in YAML

Use the document's YAML header to set an **output format** and customize it with **output options**.

```

title: "My Document"
author: "Author Name"
output:
 html_document:
 toc: TRUE

```

Indent format 2 characters,  
indent options 4 characters

| OUTPUT FORMAT                                                                                                    | CREATES                      |
|------------------------------------------------------------------------------------------------------------------|------------------------------|
| html_document                                                                                                    | .html                        |
| pdf_document*                                                                                                    | .pdf                         |
| word_document                                                                                                    | Microsoft Word (.docx)       |
| powerpoint_presentation                                                                                          | Microsoft Powerpoint (.pptx) |
| odt_document                                                                                                     | OpenDocument Text            |
| rtf_document                                                                                                     | Rich Text Format             |
| md_document                                                                                                      | Markdown                     |
| github_document                                                                                                  | Markdown for Github          |
| ioslides_presentation                                                                                            | ioslides HTML slides         |
| slidy_presentation                                                                                               | Slidy HTML slides            |
| beamer_presentation*                                                                                             | Beamer slides                |
| * Requires LaTeX, use <code>tinytex::install_tinytex()</code>                                                    |                              |
| Also see <code>flexdashboard</code> , <code>bookdown</code> , <code>distill</code> , and <code>blogdown</code> . |                              |

| IMPORTANT OPTIONS   | DESCRIPTION                                                                                                             | HTML | PDF | MS Word | MS PPT |
|---------------------|-------------------------------------------------------------------------------------------------------------------------|------|-----|---------|--------|
| anchor_sections     | Show section anchors on mouse hover (TRUE or FALSE)                                                                     | X    |     |         |        |
| citation_package    | The LaTeX package to process citations ("default", "natbib", "biblatex")                                                |      | X   |         |        |
| code_download       | Give readers an option to download the .Rmd source code (TRUE or FALSE)                                                 | X    |     |         |        |
| code_folding        | Let readers to toggle the display of R code ("none", "hide", or "show")                                                 | X    |     |         |        |
| css                 | CSS or SCSS file to use to style document (e.g. "style.css")                                                            | X    |     |         |        |
| dev                 | Graphics device to use for figure output (e.g. "png", "pdf")                                                            | X    | X   |         |        |
| df_print            | Method for printing data frames ("default", "kable", "tibble", "paged")                                                 | X    | X   | X       | X      |
| fig_caption         | Should figures be rendered with captions (TRUE or FALSE)                                                                | X    | X   | X       | X      |
| highlight           | Syntax highlighting ("tango", "pygments", "kate", "zenburn", "textmate", "arrow", "rstudio", or a custom ".theme" file) | X    | X   | X       |        |
| includes            | File of content to place in doc ("in_header", "before_body", "after_body")                                              | X    | X   |         |        |
| keep_md             | Keep the Markdown .md file generated by knitting (TRUE or FALSE)                                                        | X    | X   | X       | X      |
| keep_tex            | Keep the intermediate TEX file used to convert to PDF (TRUE or FALSE)                                                   |      | X   |         |        |
| latex_engine        | LaTeX engine for producing PDF output ("pdflatex", "xelatex", or "lualatex")                                            |      | X   |         |        |
| math_method         | Engine for rendering math in HTML ("mathjax", "katex", "r-katex", "mathml", "webtex")                                   | X    |     |         |        |
| reference_docx/_doc | docx/pptx file containing styles to copy in the output (e.g. "file.docx", "file.pptx")                                  |      |     | X       | X      |
| theme               | Theme options (see Bootswatch and Custom Themes below)                                                                  | X    |     |         |        |
| toc                 | Add a table of contents at start of document (TRUE or FALSE)                                                            | X    | X   | X       | X      |
| toc_depth           | The lowest level of headings to add to table of contents (e.g. 2, 3)                                                    | X    | X   | X       | X      |
| toc_float           | Float the table of contents to the left of the main document content (TRUE or FALSE)                                    | X    |     |         |        |

Use `?<output format>` to see all of a format's options, e.g. `?html_document`

## Render

When you render a document, rmarkdown:

1. Runs the code and embeds results and text into an .md file with knitr.
2. Converts the .md file into the output format with Pandoc.



**Save**, then **Knit** to preview the document output. The resulting HTML/PDF/MS Word/etc. document will be created and saved in the same directory as the .Rmd file.

Use `rmarkdown::render()` to render/knit in the R console. See `?render` for available options.

## Share

**Publish on Posit Connect** to share R Markdown documents securely, schedule automatic updates, and interact with parameters in real-time. [posit.co/products/enterprise/connect](https://posit.co/products/enterprise/connect).

# More Header Options

### PARAMETERS

Parameterize your documents to reuse with new inputs (e.g., data, values, etc.).

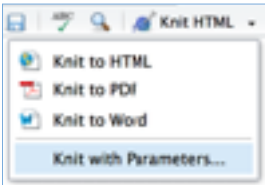
1. **Add parameters** in the header as sub-values of params.

```

params:
 state: "hawaii"

```
2. **Call parameters** in code using `params$<name>`.

```
```{r}
data <- df[, params$state]
summary(data)
```
```
3. **Set parameters** with Knit with Parameters or the params argument of `render()`.



### REUSABLE TEMPLATES

1. **Create a new package** with a `inst/rmarkdown/templates` directory.
2. **Add a folder** containing `template.yaml` (below) and `skeleton.Rmd` (template contents).

```

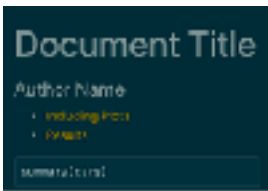
name: "My Template"

```
3. **Install** the package to access template by going to **File > New R Markdown > From Template**.

### BOOTSWATCH THEMES

Customize HTML documents with Bootswatch themes from the **bslib** package using the theme output option.

Use `bslib::bootswatch_themes()` to list available themes.



```

title: "Document Title"
author: "Author Name"
output:
 html_document:
 theme:
 bootswatch: solar

```

### CUSTOM THEMES

Customize individual HTML elements using bslib variables. Use `?bs_theme` to see more variables.

```

output:
 html_document:
 theme:
 bg: "#121212"
 fg: "#E4E4E4"
 base_font:
 google: "Prompt"

```

More on **bslib** at [pkgs.rstudio.com/bslib/](https://pkgs.rstudio.com/bslib/).

### STYLING WITH CSS AND SCSS

Add CSS and SCSS to your document by adding a path to a file with the **css** option in the YAML header.

```

title: "My Document"
author: "Author Name"
output:
 html_document:
 css: "style.css"

```

Apply CSS styling by writing HTML tags directly or:

- Use markdown to apply style attributes inline.

Bracketed Span

A `[green]{.my-color}` word.

A green word.

Fenced Div

```
::: {my-color}
```

All of these words are green.

All of these words are green.

- Use the Visual Editor. Go to **Format > Div/Span** and add CSS styling directly with Edit Attributes.

`.my-css-tag`

This is a div with some text in it.

### INTERACTIVITY

Turn your report into an interactive Shiny document in 4 steps:

1. Add **runtime: shiny** to the YAML header.
2. Call Shiny input functions to embed input objects.
3. Call Shiny render functions to embed reactive output.
4. Render with `rmarkdown::run()` or click **Run Document** in RStudio IDE.

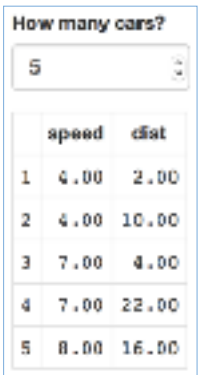
```

output: html_document
runtime: shiny

```

```
```{r, echo = FALSE}
numericInput("n",
  "How many cars?", 5)

renderTable({
  head(cars, input$n)
})
```
```



Also see Shiny Prerendered for better performance. [rmarkdown.rstudio.com/authoring\\_shiny\\_prerendered](https://rmarkdown.rstudio.com/authoring_shiny_prerendered).

Embed a complete app into your document with `shiny::shinyAppDir()`. More at [bookdown.org/yihui/rmarkdown/shiny-embedded.html](https://bookdown.org/yihui/rmarkdown/shiny-embedded.html).